

The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). At the bottom, a Windows taskbar is visible with several application icons including Chrome, Photoshop, and others. The overall image is out of focus, serving as a backdrop for the text.

C#

Level 1

Set Date & Time



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Definition

DateTime

-  `DateTime` is a .NET type used to represent a date and time.

-  It can store values ranging from a date only to a date with time and a specified kind.

Explanation

Default DateTime

-  ``new DateTime()`` creates the default ``DateTime`` value.

-  The default value is ``01/01/0001 00:00:00``.

Procedures

1. Create a `DateTime` object using `new DateTime()`.
2. Provide year, month, and day when a specific date is required.
3. Add hour, minute, and second when time is required.
4. Specify `DateTimeKind` when the time context is important.

💡 Example

```
DateTime dt1 = new DateTime();  
Console.WriteLine(dt1);  
DateTime dt2 = new DateTime(2026, 09, 03);  
Console.WriteLine(dt2);  
DateTime dt3 = new DateTime(2026, 09, 03, 13, 15, 00);  
Console.WriteLine(dt3);  
DateTime dt4 = new DateTime(  
    2026, 09, 03, 13, 15, 28,  
    DateTimeKind.Utc  
);  
Console.WriteLine(dt4);
```

➡ Result

```
01/01/0001 00:00:00  
03/09/2026 00:00:00  
03/09/2026 13:15:00  
03/09/2026 13:15:28 UTC
```

Note

- ◆ The constructor parameters follow this order:
`year, month, day, hour, minute, second`
- ◆ `DateTimeKind` has three values:
 - ▣ `Utc` → Coordinated Universal Time.
 - ▣ `Local` → Local system time.
 - ▣ `Unspecified` → No time zone information is specified.

⊘ Warning

- ◆ Month values range from `1` to `12`.
- ◆ Day values must be valid for the specified month and year.
- ◆ `new DateTime()` does not mean the current date and time; it creates the default value.



Comparison

▨ `new DateTime()`

- ✓ Creates the default date and time.
- ✓ `01/01/0001 00:00:00`.

▤ `new DateTime(year, month, day, ...)`

- ✓ Creates a specific date and optionally a specific time.
- ✓ Can also specify `DateTimeKind`.

✦ Summary

- ✓ `DateTime` represents date and time values.
- ✓ `new DateTime()` creates the default value.
- ✓ `year, month, day` create a specific date.
- ✓ `hour, minute, second` add a specific time.
- ✓ `DateTimeKind` specifies `Utc`, `Local`, or `Unspecified`.

Thank you 😊